

WiDS Kalman Filtered Trend Trader:

Project Problem Statement

Final Project

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Objective

Extend your Kalman-filtered trading strategy into a **multi-asset portfolio manager**. Instead of deciding when to trade a single asset, you will design a system that dynamically allocates capital across multiple assets using Kalman-filtered trend information and the concepts learned over the previous weeks.

The goal is to move from **single-asset trading** to **portfolio-level decision making**, while maintaining strict causality and robust backtesting.

Asset Universe

Your portfolio will consist of the following four assets:

- **Bitcoin** – High volatility, high risk
- **Nifty 50** – Indian equity market benchmark
- **Gold** – Hedge / defensive asset
- **Cash** – Risk-free holding (capital not allocated elsewhere)

You are free to model **cash** as either an explicit asset with zero return, or the absence of allocation to risky assets.

Justify your choice.

Data

You will work with daily price data from **2015-2024** for all four assets.

You may obtain the data from any reliable source, such as Yahoo Finance, Exchange-provided historical datasets, Kaggle or similar repositories.

You are responsible for:

- aligning timestamps across assets
- handling missing values
- ensuring consistent pre-processing

Kalman Filtering

You must apply **Kalman filtering** to the asset data to extract latent trend information.

- Each asset should be filtered **independently**.
- You may reuse or adapt the Kalman models developed earlier.
- You are free to define what constitutes *trend quality* (e.g., trend magnitude, velocity, stability).

Assumptions made by the Kalman filter should be clearly understood and reflected in your design choices.

Strategy Design Task

Design a **portfolio decision strategy** that, at each timestep:

- observes the current Kalman-filtered information for all assets
- decides how capital should be allocated across them

You're suggested to use a well defined rule-based logic. Implementing reinforcement learning into your strategy is very much recommended, but you're free to implement a hybrid approach too. There are **no restrictions** on the internal strategy, as long as your decisions are causal, the logic is well-defined and the system is backtestable.

Transaction Costs

To discourage excessive rebalancing, a **0.1% transaction cost** is applied to every change in allocation. This cost must be accounted for explicitly and should meaningfully affect strategy behavior.

Backtesting and Evaluation

Your portfolio strategy must be evaluated using a proper backtesting setup with:

- ensuring no lookahead bias
- decisions at time i based only on information up to i
- execution at time $i + 1$

Required Metrics

At minimum, your evaluation should include:

- Portfolio PnL and turnover.
- Annualized Return and Volatility.
- Sharpe Ratio and Maximum Drawdown.
- A benchmark against the Nifty 50 Buy-and-Hold.
- Feel free to add any of your own metrics you feel can be used to interpret and improve your idea.

Deliverables

- **Code**
 - Clean, modular Python implementation.
 - Clear separation between data processing, strategy logic, and backtesting / evaluation.
 - Please comment and explain your code properly so that it's easier for us to go through them and understand it.
- **Short Report (1–2 pages)**
 - Description of strategy logic on what you have done and thought.
 - The role of Kalman filtering in your strategy logic.
 - Summary of results and key metrics required.
 - Brief discussion on how you think this can be improved overall.

Notes, Hints, and Common Pitfalls

- **Overfitting in RL:** Training on a single market regime or time period can lead to brittle policies. Consider how your approach generalizes across different conditions.
- **Reward hacking:** Poorly designed rewards may encourage unintended behavior (e.g., excessive trading or uncontrolled risk). Always inspect resulting equity curves.
- **Model assumptions:** Kalman filters assume linear dynamics and Gaussian noise. Financial markets violate these assumptions frequently.
- **Portfolio churn:** Frequent reallocations may look profitable before costs but fail once transaction costs are applied.
- **Cash is a decision:** Choosing not to allocate capital is as meaningful as choosing an asset.

May the profit be with you.